

TEAMS THAT WORK TOGETHER, STAY TOGETHER: RESILIENCY OF ENTREPRENEURIAL TEAMS



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ABSTRACT

In this paper, we extend our previous research by investigating the stability of start-up teams and its impact on reaching operating status. Using the Panel Study of Entrepreneurial Dynamics (PSED), we analyze a nationally representative cohort of nascent entrepreneurs over three observation periods. We measure start-up team stability based on team members being added, replaced, or dropped. We also track team gender and racial composition over time. Confirming our previous findings about the role of homophily, we find active start-up teams to be relatively stable in their gender and racial composition. Further, we show that stable start-up teams are slightly more likely to achieve operational status.

INTRODUCTION

Organizations retain a significant portion of their founding characteristics as they mature. A key milestone during the founding phase is the development and institutionalization of relations among members of the startup team. Strong working relationships can lead to established core routines if team members remain and the nascent firm builds the necessary combination of relevant skills and experiences. Based on our earlier work on start-up team formation with the PSED, we know that homophily—the attraction between individuals with shared characteristics—plays a major role in the process of matching and forming these teams. Teams of nascent entrepreneurs are very homogeneous in gender, racial, and occupational backgrounds (Ruef, Aldrich, and Carter 2003). We also have preliminary evidence that racially homogenous teams may be more effective in creating an operating business, whereas gender homogeneity has no apparent impact (Aldrich et al. 2003).

Now that we understand more about the social processes that generate team formation, we can investigate whether these processes continue to guide team stability as the business emerges into an operating entity. Homophilous teams, composed of members who are similar along gender and ethnic lines, may have an advantage because of heightened social solidarity and within-group trust. However, their lack of diversity may also constitute a substantial disadvantage in uncertain environments. They may lack the flexibility to adapt to changing conditions, and might suffer from redundancy in the kinds of information sources available to them. Homophily may promote solidarity and trust, but it might also impede impersonal assessments of a team's weaknesses.

In this study, we focus on changes in team composition over time and explore whether homophily continues to be an important explanatory principle in this process by analyzing data from the Panel Study of Entrepreneurial Dynamics (PSED). Additionally, we assess whether stable teams are more likely to reach operating status than teams changing their composition.

STUDY HYPOTHESES

Homophily and the Stability of Teams

How a team is formed is inextricably linked with who becomes involved as a founder. Much of the literature begins with the assumption that because it makes instrumental sense to use a team, it therefore follows that people will be recruited for instrumental reasons. This view suggests that founders engage in an open search, using the fundamental criterion of what a potential team member can add to the venture (Mosakowski 1998). We might imagine, therefore, that a lead entrepreneur or the team's core members will not merely consider people whom they know, but also entertain the possibility of bringing strangers into the team. However, our previous findings from the PSED showed that very few strangers were part of entrepreneurial startup teams (Ruef et al. 2003).

In this study, we focus on only one aspect of team formation: homophily. In the sociological and social-psychological literatures, homophily is typically invoked as a primary explanation for who is included in groups. The "birds of a feather" hypothesis represents one of the oldest contributions of the sociological literature on differential association (Lazarsfeld and Merton 1954). The homophily principle asserts that contact occurs more frequently between people with similar characteristics than between people with dissimilar characteristics. Based on this principle, and in contrast to the principle that teams recruit first and foremost on the basis of competence, we would expect groups to be composed of people with similar characteristics. McPherson, Smith-Lovin, and Cook (2001) noted that recognition of homophily as an organizing principle of social interaction began with studies of school children's friendships in the 1920s. Lazarsfeld and Merton's study in the 1950s (1954) of friendship in two communities further boosted the salience of the principle in sociology. In the last several decades, researchers have examined discussion networks, friendship, marriage, and workplace relations, as well as other types of ties. Regardless of the type of relation examined, "the patterns of homophily are remarkably robust" (McPherson et al. 2001: 418).

McPherson, Smith-Lovin, and Cook pointed out that interest in homophily grew rapidly after WWII, sparked by two themes. First, studies of racial and ethnic segregation focused on patterns of association in schools, bases, and other public places. The pattern of same-race association was found to be quite strong. Second, researchers studying the impact of small groups on their members found that peer groups showed strong tendencies toward homophily. In principle, characteristics relevant to homophily may include social identities that are attached externally to individuals (e.g. ascribed characteristics such as gender, race, or age) or to internal states such as values, beliefs, or norms. In either case, the similarity of individuals disposes them toward a greater level of interpersonal attraction, trust, and understanding—and, consequently, greater levels of social affiliation—than would be expected among dissimilar individuals. Indeed, individuals tend to assume that others with whom they have structural bonds think as they do (Jussim and Osgood 1989; McPherson et al. 2001). This tendency toward homophily might be especially likely in organizational founding teams, which require sizable investments of time and resources (Bird 1989).

Workplace and other organizational settings tend to be highly gender segregated, creating a high level of expected homophily, based on gender. Gender homophily has been identified in a variety of task-oriented settings, including work establishments (Kalleberg et al. 1996), voluntary

organizations (McPherson and Smith-Lovin 1987), and managerial networks (Ibarra 1997). Although representative data for founding teams is sparse, researchers have found that men's business discussion networks contain few women and thus contribute to gender homogeneity (Aldrich 1999: 85-86; Carter 1994). Women's business support groups, often formed as a reaction to male dominance in entrepreneurial activities, may further enhance homophily (Aldrich 1989).

In our previous research, we found strong evidence of gender homophily in entrepreneurial founding teams. We showed that most of the apparent gender diversity in startups results from the involvement of heterosexual spouse or partner couples. When such startups were excluded from the analysis, we found a very high level of gender homophily. Entrepreneurial teams founded by people other than spouse or partner pairs display a very high degree of gender homophily. Men mostly join with other men, and women mostly with other women, thus forming gender-homogeneous teams. Were it not for teams founded around spouse/partner ties, gender-heterogeneous teams would constitute a distinct minority among all nascent firms.

We also found strong evidence of ethnic homophily, with ethnically homogeneous teams being 46 times as likely to occur as expected by chance (Ruef et al. 2003). In particular, this finding reflects the fact that homogenous minority teams are common, despite the relative rarity of minority entrepreneurs in the population as a whole. For instance, we observed 4 teams composed of four African American entrepreneurs, although only 0.02 such teams were expected under a model of random mixing. To some extent, ethnic homogeneity may be generated through kinship ties in the entrepreneurial teams. Teams involving both familial networks and ethnic diversity are extremely rare—only one case in our weighted sample matches this pattern. However, even controlling for this opportunity structure, ethnically homogeneous teams occurred at a rate that was 27 times expectations. When we examined differential levels of homophily among ethnic groups, we found that minorities had a significantly higher tendency toward homogeneity than do whites. White entrepreneurs were only 0.18 times as likely to develop mono-ethnic teams as African Americans, Asians, and non-white Hispanics.

Homophily can raise levels of perceived interpersonal trust, facilitate within-team communications, and create a sense of solidarity that may pull the team through trying times. Many studies have observed a positive relation between homophily and positive affect within a group (Hinds et al. 2000: 228-229). Homophily makes it more likely that people share a common language and knowledge base, and perhaps use shared models of their tasks and context. Homophily also facilitates greater trust and reciprocity between group members. Kamm and Nurik (1993: 18) noted that personal relationships among team members may "take top priority in the beginning, at least." Consequently, homophily should lead to more frequent communication among members and greater predictability in behavior.

H1: Homogeneous teams are more stable than heterogeneous teams in their gender and racial composition.

Based on our previous research and our literature review, we believe that homophily is a fundamental organizing principle of startup teams and that deviations from homophily are hard to sustain. Accordingly, any changes in team membership should move the team toward greater homogeneity and away from heterogeneity. For example, we expect that teams will lose members

who differ from the rest of the team in their gender or ethnicity. We also expect that new members added will most likely be of similar gender and ethnicity to members already on the team.

H2: For teams that undergo changes, homophily will dominate the transformations.

Benefits of Team Stability

In their study of how students at Carnegie Mellon University chose work group members, Hinds et al. (2000: 227) noted that reducing uncertainty has been a major theme in the literature on organizations. Uncertainty can be reduced if members feel they can exercise control over who participates in a group: "Thus, in putting together project teams and work groups, individuals will, when given the chance, construct the group that they feel is most likely to succeed." Bird (1989) implied that working together is easier if people possess complementary characteristics, rather than overlapping ones. Some research in social psychology suggests that men and women may weight these principles of attraction differently (Nevid 1984).

Some researchers have framed their arguments for the positive effects of group homogeneity in terms of the costs of diversity. Group diversity raises the costs of communication and coordination between members. Schilling and Phelps (2002) noted that several reviews of team composition have concluded that group heterogeneity, on balance, has more negative than positive effects (Bettenhausen 1991; Williams and O'Reilly 1998). For example, Dougherty's (1992) study of barriers to product innovation found that people from different departments had very different views of planning, products, and markets that were difficult to reconcile.

Membership stability thus benefits startup teams in two ways. First, the longer a team remains intact, the greater its chances of developing the solidarity and trust necessary for effective functioning. Second, stable teams benefit from improved communication as members learn to understand one another better. They will develop routines to deal with common problems and stick together when difficult situations emerge. In contrast, we expect that unstable teams will not adequately cultivate the necessary solidarity, trust, and inter-personal competencies vital to the ongoing survival of a start-up.

H3: Stable teams are more likely to reach operating status than teams with changes in composition.

DATA AND METHODS

We used data from three waves of the Panel Study of Entrepreneurial Dynamics (PSED) to explore the characteristics of newly forming entrepreneurial teams. The Entrepreneurial Research Consortium developed this project specifically to provide a basic description of the start-up process and to allow the testing of hypotheses regarding factors that effect different stages of the start-up process. The data consist of a panel of individuals in the early stages of business formation and a comparison group of non-entrepreneurs, and involves several collection phases. Individuals in the process of starting a new business are called "nascent entrepreneurs." The first phase of data collection began in 1999 and subsequent waves were collected approximately in one year intervals.

Individuals were identified through a random digit dialing telephone survey of 64,622 adults in the United States who were 18 years of age or older. Next, through a screening process, researchers identified nascent (those who were trying to start a new business alone or with others). The next research phase involved the completion of detailed phone interviews and self-completed questionnaires returned via mail. The response rate for the phone interviews was 74 percent. The phone interview (containing the data used in our analysis) had several different sections. It began with questions regarding general information about the new business and associated start-up activities. Respondents answered questions about start-up team members, the social networks in which they received support for self-employment, funding requirements, market and competition assessment, current labor force activities, work experience, migration, and family background characteristics such as birth order, parents employment history, household structure, income, and net worth.

Classification as a nascent entrepreneur required meeting several different criteria. First, individuals had to be trying to start a new business, alone or with others, and they had to be actively engaged in this during the past 12 months (6.1 percent of those screened). Second, they were expected to be owners in the new business. Third, eligible entrepreneurs were limited to those whose business efforts were in the start-up phase. Those who had already successfully started their new business did not qualify for the study. Likewise, individuals who had obtained an income from their new business that covered operating expenses and owners' salaries for three months or longer were considered to have already started their new businesses and were excluded from the study. Over 85 percent of those who qualified as nascent agreed to participate in the study. Preliminary data suggested that women would be under-represented in the final sample, and that the startup process of blacks and Hispanics might be different from that of whites. Supplemental funding from the National Science Foundation made possible an increase in the number of women participating so that they were slightly more than half of the nascent entrepreneurs in the final sample (Reynolds 1999). NSF funding also made possible an over-sampling of blacks and Hispanics so that the sample size was sufficient to test for inter-ethnic differences.

Because of the study design, we were limited in analyzing the outcome of changes in team composition for two categories: reaching operating status or actively in start-up status. We did not collect detailed information on team composition for nascent entrepreneurs who reported either an "inactive" or an "abandoned" status. Additionally, teams reporting inactive status at Wave 2 did not report any team composition information. However, if these teams participated in Wave 3 and reported either active start-up or operating status at Wave 3, team composition information was collected. Therefore, two versions of team comparisons were required at Wave 3 (Waves 1 to 3 and Waves 2 to 3).

We also have adjusted the sample by including only autonomous startups and eliminating startup efforts that were sponsored more than 50 percent by corporate entities. Some cases were also eliminated because they failed the test of not yet being cash flow positive and were judged to already be new firms. Others were taken out of the sample because of ambiguities in their responses to the interview. We have omitted solo Asian nascent entrepreneurs and all-Asian teams from the analysis, because they constituted too few cases for reliable analysis. Where Asians were part of mixed teams, we have included such teams. All results we report are for weighted data, which takes the various sampling frames into account.

RESULTS

Team Stability

We analyzed the number of teams that experienced member turnover at three observation points during the study. Team changes included the following scenarios: adding members, reducing members, adding and replacing members, and just replacing members (i.e., resulting team size stays same). For teams achieving operating status or still actively pursuing the startup, the proportion experiencing a change in membership was very low. (As we noted, we have no information on teams that dropped out of the process.) Only 10 percent of teams changed members between Wave 1 and Wave 2. Nine percent of teams changed between Waves 1 and 3, while four percent of teams changed at both Waves 2 and 3. Solo and team entrepreneurs who continued to pursue their new ventures were thus unlikely to make changes to their start-up team.

Team Composition Changes—Gender

In Table 1, we report changes between Waves 1 and 2 in terms of gender composition. The originating team composition is along the first column, while the destination composition is along the top row. The three tables correspond to the three transition opportunities. Teams without any changes in gender composition are found along the diagonal. Across all three opportunities for change, only 12 teams out of 52 teams transitioned to become mixed gender teams. Of these 12 teams, five solo teams added opposite gender team members and one all male team added a female team member. The majority of the teams thus remained gender homophilous, either by adding same gender team members or dropping opposite gender team members to become solo entrepreneurs. Because women comprise approximately 40 percent of the nascent entrepreneur population, we might have expected them to encounter constraints in attracting other women to join their new ventures. However, six out of eight solo female run start-ups added other female teammates to form all female ventures, compared to one solo female run start-up that transitioned to a mixed gender team. Similarly, six out of nine sole male run start-ups added other male teammates to form all male ventures. We also compared changes between Waves 2/3 and Waves 1/3 and found similar results of few cases transitioning into mixed gender teams (results not reported).

Team Composition Changes—Race

In Table 2, we report changes between Waves 1 and 2 in terms of racial composition. Teams remained stable in their racial composition, with very few teams located off the diagonal. Homophily again dominated the kinds of changes that occurred. Only 9 out of 52 teams transitioned to become mixed race teams. Of these 9 teams, two solo black and one Hispanic nascent entrepreneurs added team members from other racial backgrounds. Two all black teams also added team members from other racial backgrounds. Whites comprise 72 percent of the nascent entrepreneur population. No solo or all white teams added non-white team members. Seven racially homogeneous teams dropped members to become solo owners. The remaining 43 team changes were driven by homophily, either by adding individuals of the same racial background or dropping team members of different racial backgrounds. We also compared changes between Waves 2/3 and Waves 1/3 and found similar results of few cases transitioning into mixed racial teams (results not reported).

Impact of Team Composition Changes

Although active start-up teams were not likely to undergo a change in team composition, teams experiencing a change were less likely to achieve operating status by Wave 3. As we show in Table 3a, 41 percent of teams with a membership change between Waves 1 and 2 achieved operating status and 59 percent remained in an active start-up phase. In contrast, 51 percent of teams without membership changes achieved operating status. Although this association was not statistically significant, the relationship between team stability and achieving operational status was positive. In Table 3b, we divide the changes into four categories based on the timing of the change. Of the stable teams, 77 percent reached operating status by Wave 3. Of the teams experiencing membership changes, 69 percent reached operating status. These associations were statistically significant. We speculate that if change does occur, teams changing earlier or often are less likely to reaching operating status than stable teams. Teams changing later may have more time to solidify core competencies and identify key needs that new team members can address.

CONCLUSION

In our previous research on start-up teams, we found that homophily and network constraints shape how nascent entrepreneurs recruit and form teams. In this study, we extended our analysis to focus on changes in teams over time. We note **three interesting research implications** that could be pursued with the PSED data. **First**, sociological theories of entrepreneurial team formation pose a challenge to strategic theories of how entrepreneurial teams are maintained. **Strategic theories of team formation assume that economic rationality will prevail and lead to team member choices based upon the functional diversification of member characteristics.** However, our research so far has found that similarity, not difference, drives team composition. We find **little change in team composition over time, and when change does occur, it moves a team toward greater homogeneity.**

Organizational founders appear more concerned with trust and familiarity than with functional competence. Indeed, they almost **seem to be discounting competency in favor of rewarding trust.** To investigate this issue further, researchers could look more closely at the specific contributions made by team members and the terms on which they are made. At the level of specialized contributions, perhaps we will find distinctive functional competencies are relevant, even within fairly homogeneous teams.

Second, because the PSED is a longitudinal data set, it can be used to assess the long-term consequences of building entrepreneurial teams based on homophily rather than difference. Do the benefits of recruiting trustworthy others as team members outweigh the possible cost of excluding strangers? If emerging businesses benefit from strong, in-group-based ties among their members, then homophily should have a positive affect on survival. **Our results suggest that homophily inhibits change and a lack of change enhances achieving operating status. However, the differences are not great** and should be replicated in subsequent studies before being accepted as solid empirical generalizations.

Third, organizations are a significant sorting point along many dimensions of membership, especially gender and occupations. New organizations can both reproduce and challenge the existing social order, either perpetuating or disrupting tendencies already present in the culture. With-

in the United States, the composition of founding teams reflects the tendency toward gender, race, and occupational homophily. Given our results over the multiple waves of the PSED, we predict new organizations will exacerbate the already strong tendencies toward homophily in social relationships. However, we may find that homogeneous organizations ultimately do not do as well as heterogeneous ones. Founders might then be encouraged to search more widely for team members rather than fall back on the familiar and the trustworthy. Research results on entrepreneurial teams thus might have important public policy consequences as well as practical economic significance.

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Table 1
Changes in Gender Composition (Wave 1 to Wave 2)—Panel Study of Entrepreneurial Dynamics, 1998-2001. Percents by rows.

(a)

	Wave 2						%	N
	Solo Female	Solo Male	Spousal Pair	All Female	All Male	Mixed Sex		
Wave 1 Solo Female	94	0	2	3	0	1	100	72
Solo Male	0	91	0	0	6	4	100	92
Spousal Pair	0	1	97	0	0	2	100	113
All Female	8	0	0	92	0	0	100	12
All Male	0	4	0	0	95	2	100	63
Mixed Sex	0	3	0	0	0	97	100	16
Total	19	24	30	4	18	6	100	367

$\chi^2 = 1500.93$, $p = 0.000$

Note: Results are weighted

Table 2
Changes in Racial Composition (Wave 1 to Wave 2)—Panel Study of Entrepreneurial Dynamics, 1998-2001. Percents by rows.

(a)

	Wave 2							%	N
	Solo White	Solo Black	Solo Hispanic	All White	All Black	All Hispanic	Mixed Sex		
Wave 1 Solo White	98	0	0	2	0	0	0	100	110
Solo Black	0	85	0	0	8	0	7	100	34
Solo Hispanic	0	0	77	0	0	11	11	100	15
All White	0	0	0	100	0	0	0	100	128
All Black	0	4	0	0	93	0	4	100	26
All Hispanic	0	0	13	0	0	87	0	100	10
Mixed	0	4	5	0	0	0	91	100	35
Total	30	9	4	36	8	3	10	100	359

$\chi^2 = 1710.20$, $p = 0.000$

Note: Results are weighted

Table 3
Impact of Team Changes on Venture Status—Panel Study of Entrepreneurial Dynamics, 1998-2001. Changes from Wave 1 to Wave 2. Percents by rows.

		Status of Business (at Wave 2)			
		Actively		%	N
		Operating	Trying		
Change in Team Structure (Wave 1 to 2)	No	51	49	100	322
	Yes	41	59	100	35
	Total	50	50	100	357

$\chi^2 = 1.18$, $p = 0.35$ N.S.

Note: Results are weighted.

Table 4
Impact of Team Changes on Venture Status—Panel Study of Entrepreneurial Dynamics, 1998-2001. Changes Across Multiple Waves. Percents by rows.

		Status at Last Wave Interview			
		Actively		%	N
		Operating	Trying		
Change in Team Structure	No change	77	23	100	160
	Changed at Wave 3	70	30	100	16
	Changed at Wave 2, Stable at Wave 3	100	0	100	7
	Changed at both Waves 2 and 3	11	89	100	6
	Total	75	25	100	189

$\chi^2 = 18.60$, $p = 0.01$

Note: Results are weighted.